Neural Field Perceiver

Cross-Patient Speech Decoding from Intra-Operative μECoG
via Reprojection-Supervised Spatial Alignment

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Technical design document for the Neural Field Perceiver. Claims are labeled **externally established** (published), **internally observed** (our experiments), or **working hypothesis** (unverified). Structural assumptions that could be wrong are marked in red.

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1 Problem

Each patient's μECoG array is a sparse spatial sample of cortical activity during speech. The array sits on left ventral sensorimotor cortex (vSMC), and its position in MNI-152 space is known post-hoc from electrode reconstruction.

Input. For patient i with N_i electrodes at MNI positions $\{\mathbf{p}_j^{(i)}\}$, we observe high-gamma analytic amplitude (HGA, 70–150 Hz) at 200 Hz:

$$d_j^{(i)}(t) = a\big(\mathbf{p}_j^{(i)},\ t\big) + \epsilon_j^{(i)}(t) \tag{1}$$

where $a(\mathbf{p}, t)$ is the cortical HGA field and ϵ captures noise, impedance variation, and sub-electrode activity.

Output. Infer the phoneme sequence $\mathbf{y} = (y_1, y_2, y_3)$ from a 9-phoneme vocabulary — three separate 9-way classifications per non-word token (e.g., /a/, /b/, /e/).

Cross-patient challenge. Electrode j on patient i and electrode j on patient i' sample different cortical locations. Arrays are offset 15–25 mm in MNI with variable rotation. Channel indices are meaningless across patients.

Constraint	**Value**	**Impact**
Electrodes/patient	63–201	Sparse cortical sampling
Electrode spacing	∼2 mm	Near-columnar resolution
Cross-patient offset	15–25 mm	Different regions sampled
Trials/patient	46–178	Small-data regime
Utterance/patient	∼1 min	Too little for SSL
Phonemes	9 classes × 3 positions	Low-dimensional output

2 Key Idea: Shared Neural Field + Reprojection

2.1 Decomposition

Cross-patient decoding requires shared structure. We decompose each patient's cortical field:

$$a^{(i)}(\mathbf{p}, t) = f\big(T_i(\mathbf{p}),\ t;\ \mathbf{y}\big) + r^{(i)}(\mathbf{p}, t) \tag{2}$$

- f: shared neural field — the spatiotemporal pattern phoneme $\mathbf{y}$ produces in canonical cortical space.
- T_i: patient-specific spatial transform (estimated by MNI normalization, imperfect).
- $r^{(i)}$: patient-specific residual (individual neural organization, signal quality, pathology).

Cross-patient decoding works to the extent that $\|f\| \gg \|r^{(i)}\|$.

2.2 The factorization

The shared field f factorizes through a low-dimensional bottleneck:

$$d_j^{(i)}(t) = g\big(\mathbf{p}_j^{(i)},\ \mathbf{z}(t)\big) + r^{(i)}\big(\mathbf{p}_j^{(i)},\ t\big) + \epsilon_j^{(i)}(t) \tag{3}$$

where:

- $g(\mathbf{p}, \mathbf{z})$: the **somatotopic map** — given a brain position and an articulatory state, what is the neural activity? Time-invariant: lip motor cortex encodes lip state the same way at t=0 and t=500 ms. Absorbs all spatial dependence.
- $\mathbf{z}(t) \in \mathbb{R}^k$ (k≈5–8): the **articulatory state** — tongue position, lip aperture, voicing, etc. Absorbs all time and phoneme dependence.
- $\dot{\mathbf{z}}(t) = h(\mathbf{z}(t); \mathbf{y})$: dynamics on the articulatory manifold.

Space and time interact only through the k-dimensional bottleneck $\mathbf{z}(t)$. Instead of learning every (position × time × phoneme) combination, the model learns a spatial basis g and temporal dynamics h composed through $\mathbf{z}$. This is what makes the problem tractable with 63–201 electrodes.

The somatotopic map is static. g does not change during a trial. Cross-patient alignment is therefore a static geometric problem: different patients have electrodes at different positions sampling the same map. The temporal dynamics $\mathbf{z}(t)$ are shared across patients — the same phoneme produces similar trajectories regardless of who speaks it.

2.3 The 3D vision analogy

This structure maps to multi-view 3D reconstruction:

3D Reconstruction	**Neural Field Decoding**
3D body model	Shared somatotopic map $g(\mathbf{p}, \mathbf{z})$
Pose parameters	Articulatory state $\mathbf{z}(t)$
Camera extrinsics	MNI offset $\Delta^{(i)}$, rotation $\omega^{(i)}$
Camera intrinsics	Per-patient gain/bias
2D keypoints	Electrode observations $d_j^{(i)}(t)$
Reprojection loss $\|\pi(V) - x\|^2$	**Reprojection loss** $\|g(\mathbf{p}, \hat{\mathbf{z}}) - d\|^2$

Reprojection loss: if your latent state estimate is correct, you should predict what every electrode observes. Self-supervised but geometrically grounded — unlike JEPA/VICReg/masked reconstruction, the target is physically meaningful.

Where the analogy breaks. (1) Each patient is a different "scene," not a different camera on the same scene — the shared structure is weaker than photometric consistency. (2) We have 63–201 electrodes, not 65K pixels — the reprojection signal per frame is thin. (3) The "scene" changes every frame. These mean reprojection is a useful regularizer, not the primary signal. The supervised classification loss remains primary.

2.4 Evidence

Externally established. Per-patient input layers + shared backbone transfers in related settings: Singh [1] (N=25 sEEG, PER 0.49 group vs. 0.57 individual); Willett [2] (per-day affine + shared GRU). Articulatory somatotopy in vSMC is reproducible [3, 4]. Motor manifold structure is preserved across individuals [5, 6]. MNI coordinates help cross-patient ECoG transfer: Chen [7] (SwinTW, PCC 0.765 on unseen patients).[1]

[1] Standard ECoG (10 mm spacing), not μECoG (2 mm). Coordinates may encode global position but not within-array relationships at our resolution.

Internally observed. LOPO achieves PER = 0.762 on S14 (frozen backbone, grouped-by-token CV) — above chance (0.89) but worse than per-patient (0.737). 55 experiments converge to PER 0.750–0.780, suggesting a systematic barrier (data scarcity, alignment, or both). Articulatory bottleneck head improves LOPO decoding, consistent with f organized along articulatory dimensions.

3 Architecture

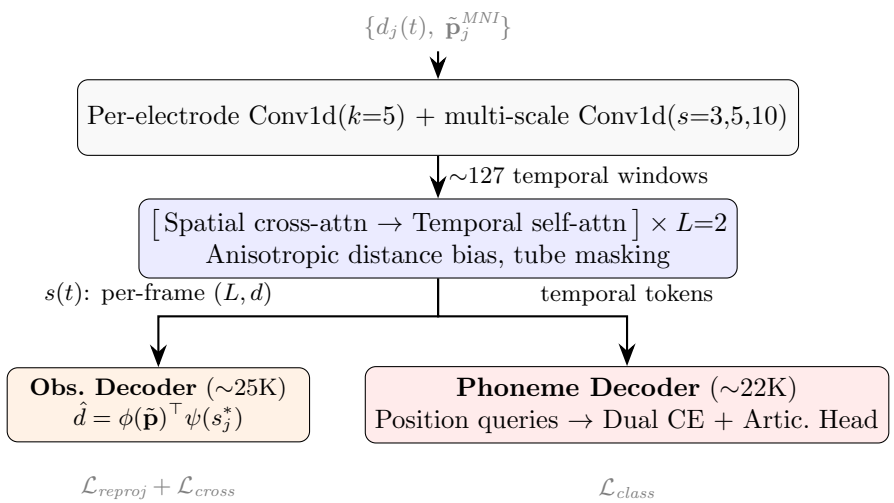
The model prediction is:

$$\boxed{\hat{d}_j^{(i)}(t) = g_\theta\big(\tilde{\mathbf{p}}_j^{(i)},\ s^{(i)}(t),\ e^{(i)}\big)} \tag{4}$$

where $\tilde{\mathbf{p}}_j^{(i)}$ is the corrected electrode position, $s^{(i)}(t) \in \mathbb{R}^{L\times d}$ is the spatial representation (L=16 virtual electrodes), and $e^{(i)}$ is a patient embedding. This is deterministic; the observation model $d = \hat{d} + \epsilon$ separates prediction from noise.

Unlike the idealized Eq. 3, the implemented model conditions on patient identity through $e^{(i)}$ and corrected positions $\tilde{\mathbf{p}}_j^{(i)}$ — a deliberate relaxation to handle the patient-specific residual $r^{(i)}$.

3.1 Overview



$$\boxed{\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda_1\mathcal{L}_{\text{reproj}} + \lambda_2\mathcal{L}_{\text{cross}}, \qquad \lambda_1\text{: } 0.5\to 0.1, \quad \lambda_2\text{: } 0.3\to 0.05} \tag{5}$$

Factorized interleaved processing replaces strict spatial→temporal separation. Each layer alternates spatial cross-attention (virtual electrodes aggregate from real electrodes within a time window) and temporal self-attention (each virtual electrode attends across time windows). The spatial distance bias is **static** — the same at every time step — respecting the time-invariance of g (Section 2.2). Temporal attention captures dynamics in $\mathbf{z}(t)$. Interleaving lets temporal context denoise spatial estimates at every layer.

Two representations:

- $s(t) \in \mathbb{R}^{L\times d}$: virtual electrode outputs after spatial cross-attention. Used by the observation decoder for reprojection.

- $\mathbf{z}_p \in \mathbb{R}^d$: per-phoneme-position readout via learned query cross-attention over temporal tokens. Used for classification.

3.2 Electrode Tokens

For electrode j on patient i at time t:

$$\text{token}_j = \underbrace{\text{Linear}(d_j \to d)}_{\text{activity}} + \underbrace{\text{Linear}\big(\gamma(\tilde{\mathbf{p}}_j) \to d\big)}_{\text{Fourier MNI PE}} + \underbrace{e^{(i)}}_{\text{patient}} \tag{6}$$

where $\tilde{\mathbf{p}}_j = R(\omega^{(i)}) \cdot \mathbf{p}_j + \Delta^{(i)}$ is the corrected MNI position.

Fourier positional encoding. Following NeRF [17], MNI coordinates pass through multi-scale sinusoidal features before projection:

$$\gamma(\mathbf{p}) = \big[\sin(2^0\pi\mathbf{p}),\ \cos(2^0\pi\mathbf{p}),\ \ldots,\ \sin(2^{F-1}\pi\mathbf{p}),\ \cos(2^{F-1}\pi\mathbf{p})\big] \in \mathbb{R}^{6F} \tag{7}$$

with F=4 frequency levels. This captures spatial structure at multiple scales: coarse (∼15 mm articulatory region boundaries), medium (∼5 mm within-region gradients), and fine (∼2 mm electrode-scale variation). A linear projection Linear($24 \to d$) maps to token dimension.[2]

Per-patient parameters ($6 + d$ per patient):

- $\Delta^{(i)} \in \mathbb{R}^3$: translation (init $\mathbf{0}$). Corrects placement offset.
- $\omega^{(i)} \in \mathbb{R}^3$: axis-angle rotation (init $\mathbf{0}$, Rodrigues formula). Corrects orientation.
- $e^{(i)} \in \mathbb{R}^d$: patient embedding (init $\sim \mathcal{N}(0, 0.02)$). Absorbs residual $r^{(i)}$.

Assumption: $e^{(i)}$ is added uniformly to all tokens — the encoder cannot make position-dependent patient corrections. Only the decoder can (via MLP($\tilde{\mathbf{p}}, e$)). For tumors with spatially non-uniform distortion, this is a known limitation.

3.3 Temporal Preprocessing

Per-electrode denoising. Conv1d(k=5, stride=1) on each electrode's time series provides ∼25 ms temporal smoothing.

Multi-scale tokenization. Three parallel Conv1d branches at strides 3, 5, 10 (∼67, 40, 20 Hz) on each denoised electrode time series. These capture fast articulatory transitions (∼15 ms), syllable dynamics (∼25 ms), and phoneme-level patterns (∼50 ms). Tokenization uses the production window $[0, T_{\text{utt}}]$ (≈1.0 s, 200 samples at 200 Hz): stride 3 yields 66 tokens, stride 5 yields 40, stride 10 yields 20, totaling ≈126 temporal tokens per electrode. Concatenated with learnable per-scale embeddings + sinusoidal temporal PE (encoding each token's center time in seconds).

3.4 Virtual Electrodes

$L = 16$ learned query vectors at learned MNI positions. Fixed model parameters, the same for every patient and time step. During spatial cross-attention, each virtual electrode attends to all real electrodes, weighted by spatial proximity.

[2] NeRF uses F=10. We use F=4 to avoid overfitting with ∼200 electrode positions across ∼20 patients. The maximum frequency resolves ∼1 mm — below electrode spacing.

After cross-attention, virtual electrode ℓ holds a d-dimensional summary of cortical activity near its position. The output $s(t) \in \mathbb{R}^{L \times d}$ is a spatially-indexed representation in a shared frame — every patient maps to the same 16 positions regardless of electrode count or grid layout.

Atlas initialization. Virtual electrode positions are initialized at Brainnetome atlas [19] centroids of speech motor subregions: lip M1, tongue M1, jaw M1, laryngeal M1, lip S1, tongue S1, ventral premotor, supplementary motor area, and adjacent somatosensory zones. This gives anatomically meaningful starting positions so backprop only needs to fine-tune, not discover cortical organization from scratch. Repulsive regularization $\mathcal{L}_{\text{repel}} = \sum_{i \neq j} \max(0, r_{\min} - \|\text{v_pos}_i - \text{v_pos}_j\|)$ prevents collapse.

Why 16. 63–201 electrodes cover $\sim 20 \times 30$ mm. 16 virtual electrodes gives ~ 5 mm spacing, roughly matching somatotopic scale. Ablation A4 tests $\{4, 8, 16, 32\}$.

Coverage gaps. Virtual electrodes in uncovered cortical regions receive only weak, distant attention and will drift toward covered regions or produce noise. The union of all patients' coverage is larger than any single patient's — more patients and cross-task pooling expand the coverage envelope. No architecture fixes missing data.

3.5 Interleaved Spatiotemporal Blocks ($\times L$=2)

Each block alternates two attention operations:

(a) Spatial cross-attention (per temporal window). Virtual electrodes (Q) attend to real electrode tokens (K, V) within each time window. Attention logits use **QK-Norm + anisotropic distance bias**:

$$A_{ij} = \tau_h \cdot \frac{Q_i \cdot K_j}{\|Q_i\| \, \|K_j\|} \; - \; (\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_j)^\top \Sigma_i^{-1} (\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_j) \tag{8}$$

where $\tau_h > 0$ is a learnable temperature per head (init $\sqrt{d_h}$, clamped ≤ 100), $\bar{\mathbf{p}}$ denotes standardized MNI coordinates, and $\Sigma_i = L_i L_i^\top$ is a per-virtual-electrode covariance (Cholesky-parameterized, 6 params each, init αI). Normalizing Q and K to unit vectors keeps the content term bounded, so the distance bias has a consistent effect throughout training regardless of embedding norms — following SwinV2 [7] and ViT-22B. Each virtual electrode learns the **shape** of its spatial receptive field: an elongated lip region, a compact tongue area, etc. 96 + 4 params (16×6 covariances + 4 temperatures). 4-head attention.

The distance bias is **static**: Σ_i depends only on virtual electrode identity, not on time or input content. The spatial geometry of attention is the same at every time step, consistent with g being time-invariant. What changes with time are the **values** (electrode activity), not the spatial weighting structure.

(b) Temporal self-attention (per virtual electrode). Each of the 16 virtual electrodes attends across all $\sim$127 temporal windows. 4 heads, sinusoidal temporal PE + per-scale embeddings. This captures articulatory dynamics $h(\mathbf{z}; \mathbf{y})$.

Optional 1-layer spatial self-attention between virtual electrodes within each time window (captures inter-zone co-articulation, e.g., simultaneous lip rounding and tongue backing).

Block internals. Each sub-layer uses Pre-LayerNorm: $\text{out} = x + \text{SubLayer}(\text{LN}(x))$. The FFN after each attention operation is SwiGLU [24]:

$$\text{FFN}(x) = \big(xW_1 \odot \text{SiLU}(xW_{\text{gate}})\big)W_2, \qquad W_1, W_{\text{gate}} \in \mathbb{R}^{d \times d_{\text{ff}}}, \; W_2 \in \mathbb{R}^{d_{\text{ff}} \times d} \tag{9}$$

with $d_{\text{ff}} = \lfloor 8d/3 \rfloor$, matching the parameter count of a standard $4d$ ReLU FFN. Dropout p=0.2 on attention weights and FFN output. DropPath p=0.1 on the residual stream. Each block thus contains: LN → spatial cross-attn → LN → FFN → LN → temporal self-attn → LN → FFN.

Tube masking. Instead of masking electrodes independently per frame, mask each selected electrode across a contiguous time window of 5–10 frames (25–50 ms). Masking is **scale-consistent**: when electrode j is masked at time t, it is masked at all three temporal resolutions simultaneously. Otherwise the model trivially reconstructs masked information from a finer or coarser scale — the somatotopic map $g(\mathbf{p}, \mathbf{z})$ is the same physical function at every temporal resolution. This forces interpolation in **both** space (from nearby electrodes at the same time) **and** time (from the same electrode at adjacent windows), exercising both types of shared structure.

Masking curriculum. Start at 30%, anneal to 70% over training. Low masking early helps the spatial encoder learn basic distance-activity relationships. High masking later forces robust spatial interpolation. The somatotopic map is spatially smooth (∼5 mm scale), so 70% masking is physically feasible.

Spatial readout. Attention pooling (not MaxPool) preserves the somatotopic decomposition:

$$s_{\text{pool}}(t) = \sum_{i=1}^{L} w_i \cdot \text{out}_i(t), \qquad w_i = \text{softmax}\big(\mathbf{q}_{\text{read}}^{\top} \text{out}_i(t)/\sqrt{d}\big) \tag{10}$$

Per-position readout via learned queries. Three learned query vectors $\mathbf{q}_p \in \mathbb{R}^d$ ($p \in \{1, 2, 3\}$) cross-attend to the full spatially-pooled temporal token sequence:

$$\mathbf{z}_p = \text{CrossAttn}\big(\mathbf{q}_p,\ K{=}\{s_{\text{pool}}(t)\}_t,\ V{=}\{s_{\text{pool}}(t)\}_t\big) \tag{11}$$

Each query discovers where its phoneme's information concentrates in the temporal sequence, accounting for variable phoneme durations, coarticulation across boundaries, and CVC/VCV structural asymmetry (VCV: 0.68 s vs. CVC: 0.58 s [10]). Inspired by DETR's object queries [23], which learn to attend to task-relevant regions without hard spatial partitioning. Ablation A13 compares against fixed equal-window averaging. Global readout z_{global} = mean(all temporal tokens) for k-NN.

3.6 Why Perceiver, not ViT or point-cloud methods?

ViT requires a fixed regular grid. Our patients have different grid sizes (8×16, 12×22) at different MNI positions. Cannot use.

Point-cloud methods (PointNet [15], Point Transformer [16]) handle irregular 3D point sets and are legitimate alternatives. Perceiver cross-attention is simpler for our case: shared query vectors directly provide the fixed-size shared output we need, and our 63–201 points are too small for hierarchical architectures. Worth testing as an ablation.

3.7 Observation Decoder (∼25K params)

The forward model — predicts what each electrode should observe given the latent state. Structurally, this is a DeepONet [18]: a branch network encodes the latent field, a trunk network encodes spatial position, and they interact through a low-rank inner product. The neural operator literature validates this as the right inductive bias for sparse field reconstruction.

Position-weighted aggregation. Each electrode j gets a local latent by weighting virtual electrode outputs by spatial proximity:

$$s_j^*(t) = \sum_{\ell=1}^{L} a_{j\ell} \cdot \mathrm{out}_\ell(t), \qquad a_{j\ell} = \frac{\exp\big(-\alpha_{\mathrm{dec}} \cdot \|\bar{\tilde{\mathbf{p}}}_j - \mathrm{v_\bar{p}os}_\ell\|^2\big)}{\sum_m \exp\big(-\alpha_{\mathrm{dec}} \cdot \|\bar{\tilde{\mathbf{p}}}_j - \mathrm{v_\bar{p}os}_m\|^2\big)} \tag{12}$$

Bilinear prediction.

$$\hat{d}_j^{(i)}(t) = \phi(\tilde{\mathbf{p}}_j,\ e^{(i)})^\top\ \psi\big(s_j^*(t)\big) \tag{13}$$

$\phi = \mathrm{MLP}_{\mathrm{spatial}}(\tilde{\mathbf{p}}, e) \in \mathbb{R}^k$: spatial tuning (trunk). $\psi = \mathrm{MLP}_{\mathrm{latent}}(s_j^*) \in \mathbb{R}^k$: latent activation (branch). k=32.

Unlike the encoder (where $e^{(i)}$ is a global bias), the decoder's $\phi(\tilde{\mathbf{p}}, e)$ learns position-dependent patient effects — partially modeling the nonlinear residual $\delta_i(\mathbf{p})$. Ablation A9 tests a full MLP fallback.

Within-patient reprojection loss.

$$\mathcal{L}_{\mathrm{reproj}} = \frac{1}{N_i T} \sum_{t,j} \big\| d_j(t) - \hat{d}_j(t) \big\|^2 \tag{14}$$

Computed over all electrodes including masked ones (the model must interpolate).

Cross-patient reprojection loss. During multi-patient training, when patients A and B have trials with the same phoneme sequence, the shared observation function g should explain **both** patients' observations from either patient's latent state. For each matched pair:

$$\mathcal{L}_{\mathrm{cross}} = \frac{1}{N_B} \sum_{j=1}^{N_B} \big\| \bar{d}_j^{(B)} - g_\theta(\tilde{\mathbf{p}}_j^{(B)},\ s_A^{*,(B)}(t),\ e^{(B)}) \big\|^2 \tag{15}$$

where $\bar{d}_j^{(B)}$ is patient B's position-averaged observation (averaged over the matching phoneme window to handle timing differences), and $s_A^{*,(B)}$ aggregates patient A's virtual electrode outputs at patient B's electrode positions.

This is the multi-view analogy made literal: if A and B produce /abe/, the latent state extracted from A's electrodes should predict B's observations when conditioned on B's positions and identity. It directly forces the spatial encoder to learn patient-invariant representations.

Caution: Cross-patient reprojection is noisier than within-patient because (1) the patient residuals $r^{(A)} \neq r^{(B)}$ mean a perfect g still leaves unexplained variance, and (2) position-averaging over a phoneme window blurs temporal dynamics. Weight $\lambda_2 < \lambda_1$ and anneal aggressively. If $\mathcal{L}_{\mathrm{cross}}$ competes with $\mathcal{L}_{\mathrm{class}}$, the λ_2=0 fallback drops it entirely.

Optional: Beta-NLL. Replace MSE with learned per-electrode variance to downweight noisy electrodes:

$$\mathcal{L}_{\mathrm{reproj}}^{\mathrm{UQ}} = \frac{1}{2}\Big(\log \hat{\sigma}_j^2 + \frac{(d_j - \hat{d}_j)^2}{\hat{\sigma}_j^2}\Big) \cdot \big(\hat{\sigma}_j^2\big)_{\mathrm{sg}}^{0.5} \tag{16}$$

3.8 Phoneme Decoder (∼5K params)

Dual head per position $p \in \{1, 2, 3\}$:

$$\text{logits}_p = \frac{1}{2}\Big(\underbrace{\text{Linear}(z_p \to 9)}_{\text{CE head}} + \underbrace{\tau \cdot \text{normalize}(\text{Linear}(z_p \to 15)) \cdot A^\top}_{\text{articulatory head}}\Big) \tag{17}$$

$A \in \mathbb{R}^{9\times 15}$: fixed articulatory feature matrix. τ: learnable temperature. Separate heads per position. $\mathcal{L}_{\text{class}} = \frac{1}{3}\sum_p \text{FocalCE}(\text{logits}_p, y_p;\ \gamma{=}2, \text{LS}{=}0.1)$ with mixup (α=0.2).

3.9 Parameter Budget

Module	**Params**	**Shared?**	**Role**
Interleaved blocks (×2)	∼160K	Shared	Spatial inversion + dynamics
incl. anisotropic $\Sigma_i + \tau_h$	100	Shared	Receptive field shape + QK temp
incl. virtual positions	48	Shared	Atlas-initialized MNI coords
Position queries + cross-attn	∼17K	Shared	Learned phoneme readout
Observation decoder	∼25K	Shared	Reprojection forward model
Phoneme dual heads (×3)	∼5K	Shared	Classification
Multi-scale Conv1d	∼12K	Shared	Temporal tokenization
$\Delta^{(i)} + \omega^{(i)} + e^{(i)}$	70/pt	Per-patient	Registration + identity
Total shared	**∼219K**		
Grand total (10 pts)	**∼220K**		(cf. current: 122K)

4 Training and Evaluation

4.1 Phase 1: Source training

9 source patients, ∼1315 trials. Each batch:

1. Construct tokens with Fourier MNI PE. Apply tube masking (curriculum 30→70%).
2. Interleaved blocks: spatial cross-attn → temporal self-attn $\times L$.
3. Within-patient reprojection on all electrodes (including masked) → $\mathcal{L}_{\text{reproj}}$.
4. Cross-patient reprojection on same-token trial pairs → $\mathcal{L}_{\text{cross}}$.
5. Per-position readout → dual head → $\mathcal{L}_{\text{class}}$.
6. $\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda_1\mathcal{L}_{\text{reproj}} + \lambda_2\mathcal{L}_{\text{cross}}$.

AdamW with differential LR: shared params 10^{-3}; geometric params (Δ, ω, virtual positions, Σ) $3{\times}10^{-4}$; patient e at 10^{-3}. Linear warmup (20 epochs) → cosine decay. Weight decay 10^{-4}, grad clip 5.0, batch 16. Early stopping on held-out 20% source validation, patience 7.

4.2 Phase 2: Target adaptation (LP-FT)

Per fold of 5-fold grouped-by-token CV on target patient:

1. **Linear probe.** Freeze encoder, train decoder + $\Delta^{(\text{target})} + \omega^{(\text{target})} + e^{(\text{target})}$. 150 epochs. Best checkpoint by inner validation (20% held out from training fold).

2. **Fine-tune.** Unfreeze encoder at 0.1× LR. 100 epochs, patience 7.

4.3 Phase 3 (optional): Transductive reprojection pre-training

Pre-train spatial encoder + decoder using $\mathcal{L}_{\text{reproj}}$ only. If target patient's unlabeled data is included, this is transductive and must be reported separately from strict LOPO.

4.4 Evaluation regimes

1. **Zero-shot.** Phase 1 only. $\Delta = \mathbf{0}$, $\omega = \mathbf{0}$, $e = \text{mean}(e^{(\text{source})})$, all frozen. Strongest claim, likely worst PER.
2. **LP-FT (primary).** Phase 1 → Phase 2. Matches clinical calibration scenario.
3. **Transductive.** Phase 3 → Phase 1 → Phase 2. Reported separately.

Metrics. PER (edit distance), grouped-by-token 5-fold CV. Full recipe: weighted k-NN (k=10, cosine), TTA 16, dual head, source k-NN at 0.5× weight. Conformal prediction sets as interpretability output.

5 What Our Experiments Show

5.1 Evaluation recipe matters more than architecture

Component added	Category	PER	Δ
Baseline (CE, mean-pool)	–	0.872	–
+ Label smoothing 0.1	Training	0.831	−4.1 pp
+ Per-position heads + dropout	Training	0.819	−1.2 pp
+ Mixup α=0.2	Training	0.802	−1.7 pp
+ Weighted k-NN (k=10)	Evaluation	0.781	−2.1 pp
+ TTA (16 copies)	Evaluation	0.767	−1.4 pp
+ Articulatory + CE dual head	Evaluation	0.737	−3.0 pp

Cumulative, not independent. Total: −13.5 pp. All on S14, grouped-by-token CV. Per-fold noise is ±0.05 PER, so improvements < 2 pp may not replicate.

5.2 Multi-scale temporal is the best architectural improvement

Stride 3+5+10 gave PER 0.766 → 0.750 (−1.6 pp). Validated on BiGRU; needs re-validation on transformer.

5.3 The ∼0.76 convergence

55 LOPO experiments converge to PER 0.750–0.780. Not attributable to any single bottleneck: S14 is no longer truly held out (55+ architectures evaluated); ∼30 samples/fold give ±0.05 noise; S14 has only 153 trials; spatial alignment is one factor among several.

5.4 SSL failed at this data scale

All SSL methods near chance (PER 0.85–0.91). JEPA + LP-FT: 0.797. With ∼1 min utterance/patient, SSL cannot learn useful representations from HGA.

6 Ablation Plan

ID	Ablation	Question
A1	MNI PE present/absent	**Central**: do coordinates help?
	a. Current Conv2d+BiGRU baseline	$\rightarrow$ 0.762 (LP-FT)
	b. Perceiver, random positions	$\rightarrow$ architecture benefit
	c. Perceiver + Fourier MNI PE	$\rightarrow$ MNI benefit
	d. (c) + reprojection loss	$\rightarrow$ reprojection benefit
	e. (d) + cross-patient reproj.	$\rightarrow$ cross-patient benefit
A2	Population-level (all patients)	Is the 0.76 wall S14-specific?
A3	Per-patient params	Δ only / $+\omega$ / $+e$ / pos-dep e
A4	Virtual electrode count	$L \in \{4, 8, 16, 32\}$
A5	Reprojection weight	$\lambda_1, \lambda_2 \in \{0, 0.01, 0.1, 0.5\}$
A6	Cross-task pooling	+ Duraivel 2025 data
A7	Reprojection loss variant	Beta-NLL vs. MSE
A8	Spatial attention	Isotropic α vs. anisotropic Σ_i
A9	Bilinear vs. MLP decoder	$\phi^\top \psi$ vs. MLP($[\tilde{\mathbf{p}}; e; s_j^*]$)
A10	Interleaved vs. separated	Interleaved blocks vs. strict spatial$\rightarrow$temporal
A11	Masking strategy	Tube vs. per-frame; masking ratio 30–80%
A12	Fourier PE vs. linear PE	$\gamma(\mathbf{p})$ vs. Linear($\mathbf{p} \rightarrow d$)
A13	Position readout	Learned queries vs. fixed equal-window averaging

7 Priorities and Risks

7.1 Priority ordering

The document's own evidence suggests:

1. **Cross-task data pooling (A6).** Lowest risk. Same phonemes, overlapping patients [10]. Directly addresses data scarcity. Works with current pipeline and new architecture.
2. **Population-level evaluation (A2).** Pure methodology fix. Required to distinguish S14-specific effects from universal trends. Must be the primary success criterion.
3. **MNI coordinate architecture (A1, this document).** Highest novelty, highest risk. The hypothesis that spatial alignment is the binding constraint is plausible but unvalidated — it has not been isolated from data scarcity or measurement noise.

7.2 Success criteria

Primary: Mean PER across all patients improves over current LOPO baseline (≥ 3 seeds, LP-FT regime). MNI benefit isolated by paired comparisons: A1c vs. A1b (MNI value) and A1d vs. A1c (reprojection value).

Interpretability: Virtual electrode positions cluster over known speech motor regions. Spatial tuning vectors $\phi(\mathbf{p})$ recover articulatory somatotopy. Learned $\Delta^{(i)}$ and $\omega^{(i)}$ correlate with known registration quality.

Risk	Severity	Mitigation
MNI coordinates unavailable	High	Fallback to random-init positions
MNI alignment too imprecise	High	Ablation A1; learnable α adapts
Reprojection competes with classification	Med	λ ablation A5; λ=0 fallback
Perceiver overfits (200K, 1300 trials)	Med	Masking; early stopping; reproj. reg.
Bilinear separability too restrictive	Med	Ablation A9: MLP fallback
Virtual electrodes collapse	Low	Atlas init; repulsive regularization

7.3 Risks

8 Available Data

Cohort	Patients	Trials	MNI?	Usage
PS source (Spalding 8)	9 patients	∼1162	Expected*	Phase 1
PS target (pilot)	S14	153	Expected*	Phase 2 eval
Additional PS	S16,S32,S36,S57	∼400	Expected*	Future source
Cross-task (Duraivel 2025)	3 overlapping pts	156–208/pt	Expected*	Priority 1
Lexical task	13 patients	Varies	Expected*	Future source
Total unique	**∼25 patients**			

*MNI coordinates expected from BioImage Suite / Brainlab outputs on Box. Not yet extracted. Fallback: scaled grid positions (retains within-patient structure, loses cross-patient alignment). S32 = S36 (same patient recorded twice; 11 unique PS patients).

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